

**Vishaali B 251001304-2025-IT** ▾**V2****Started on** Friday, 21 August 2026, 1:47 PM**State** Finished**Completed on** Friday, 21 August 2026, 2:04 PM**Time taken** 16 mins 18 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      int counter=0;
7      counter++;
8      scanf("%d", &n);
9      int c=0;
10     counter++;
11     for (int i = n / 2; i < n; i++)
12     {
13         counter++;
14         counter++;
15         for (int j = 1; j < n; j = 2 * j)
16         {
17             counter++;
18             counter++;
19             for (int k = 1; k < n; k = k * 2)
20             {
21                 counter++;
22
23
24                 c++;
25                 counter++;
26             }
27
28
29         }
30
31     }
32
33     printf("%d",counter);
34
35     return 0;
36 }
```

	Input	Expected	Got	
✓	4	30	30	✓

	Input	Expected	Got	
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)